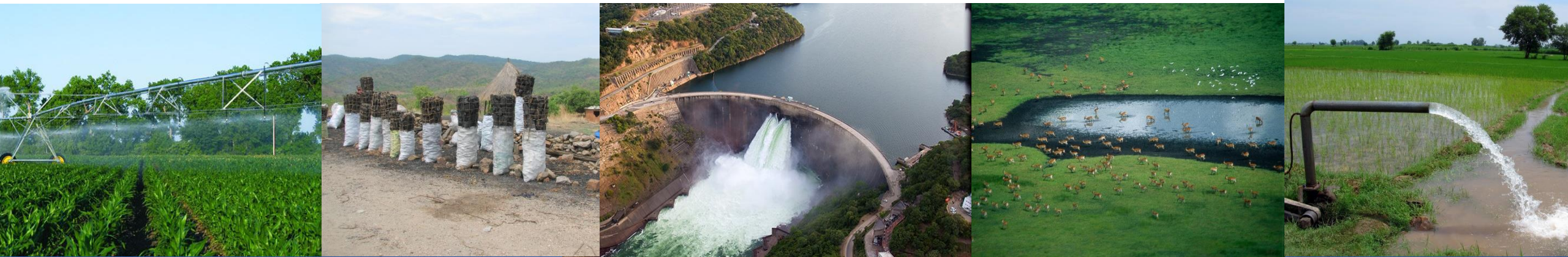


Exercise 4: Correction of the River Network

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Water Research Group



River Network – Exercise 4

Correction of the River Network

1. Routing in CWatM
2. River network
3. River network representation in CWatM
4. Tool to correct river network (ArcGIS)



River Network – Exercise 4

Dynamic vs kinematic Wave for River Routing

The **kinematic wave** and **dynamic wave** are concepts in hydrology and fluid mechanics, particularly in modeling the movement of water over land or in open channels. They are part of the family of shallow water wave equations

Dynamic wave: The dynamic wave represents the full set of shallow water equations (Saint-Venant equations), considering all forces acting on the flow, including gravity, friction, inertia, and pressure

Kinematic wave: The kinematic wave is a simplified model of flow that assumes the flow is driven primarily by gravity, with negligible effects from pressure and inertia

Key Differences

Aspect	Kinematic Wave	Dynamic Wave
Forces Considered	Gravity and friction.	Gravity, friction, inertia, and pressure.
Computational Demand	Low.	High.
Flow Behavior	Gradual changes, uniform flow approximations.	Rapid changes, backwater, and complex flow.
Equation Complexity	Simplified momentum equation.	Full momentum equation.
Applications	Watershed and surface runoff modeling.	Flood, dam-break, and unsteady flow analysis.

ChatGPT, 2024

River Network – Exercise 4

River Routing in CWatM

Runoff (surface runoff + baseflow) is routed through the river network using a 4-point implicit finite-difference solution of the kinematic wave equations (Chow, 1988). The basic equations used are the equations of continuity and momentum.

The continuity equation is:

$$\frac{\partial Q_{sr}}{\partial x} + \frac{\partial A_{sr}}{\partial t} = q_{sr}$$

where Q_{sr} is the surface runoff [m^3], A_{sr} is the cross-sectional area of the flow [m^2] and q_{sr} is the amount of lateral inflow per unit flow length [m^2s].

The momentum equation is defined as:

$$\rho \cdot g \cdot A_{sr} \cdot (S_0 - S_f) = 0$$

where ρ is the density of the flow [$kg\ m^3$], g is the gravity acceleration [$m\ s^2$], S_0 is the topographical gradient and S_f is the friction gradient. From the momentum equation it follows that $S_0 = S_f$, which means that for the kinematic wave equations it is assumed that the water surface is parallel to the topographical surface.

River Network – Exercise 4

River Routing in CWatM

What data we need for kinematic wave in CWatM?

River network:	connections of the grid network
Channel width:	average width of the main river in each gridcell
Chanbnkf:	average bankfull channel depth
Changradient:	average slope of the main river
Chanleng:	Length of the main river in a grid cell (inc. meandering)
Chanman:	Manning's roughness

From river network and cell area you can calculate upstream area, (sub) watersheds

The last 5 maps are calculated based on river network, upstream area, DEM, high resolution network

The most import element here is the connection of grid-cells on a 1 arcmin raster

River Network – Exercise 4

River Routing in CWatM

Runoff (surface runoff + baseflow) is routed through the river network using a 4-point implicit finite-difference solution of the kinematic wave equations (Chow, 1988). In CWatM this is done in C++ for faster performance.

(Deterministic) D8 River Network

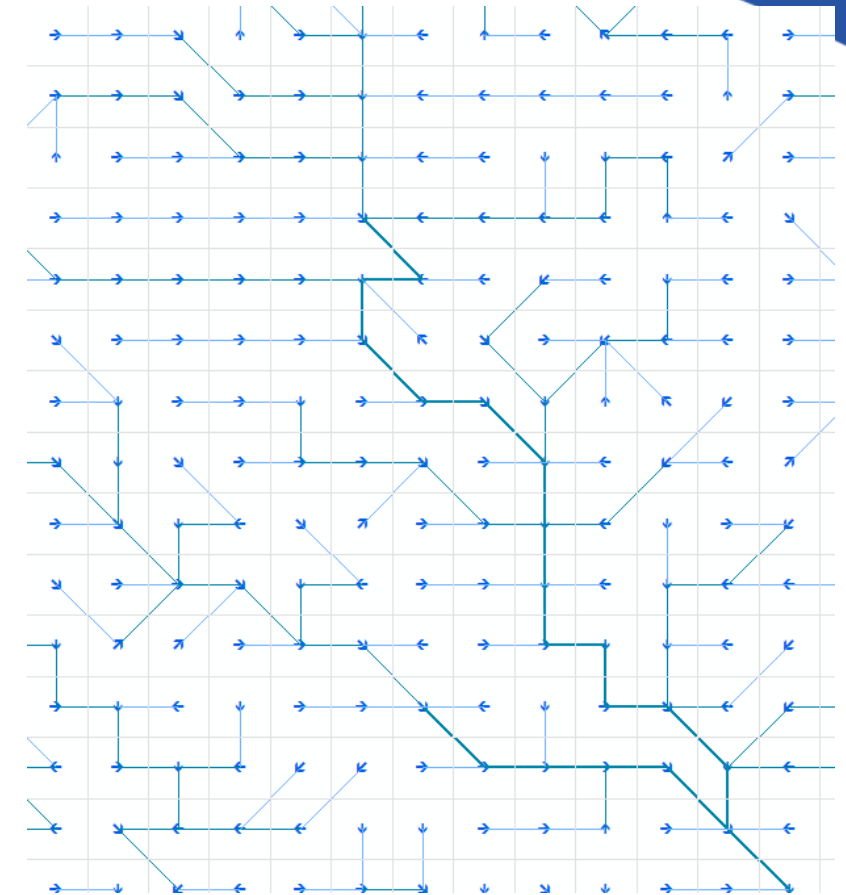
- Water flow is modeled to move from a cell to one of its eight neighboring cells (north, northeast, east, southeast, south, southwest, west, northwest).
- The direction is determined based on the steepest downslope gradient.

In CWatM Danube the river network comes from:

DEM: Merit – 3 arcsec

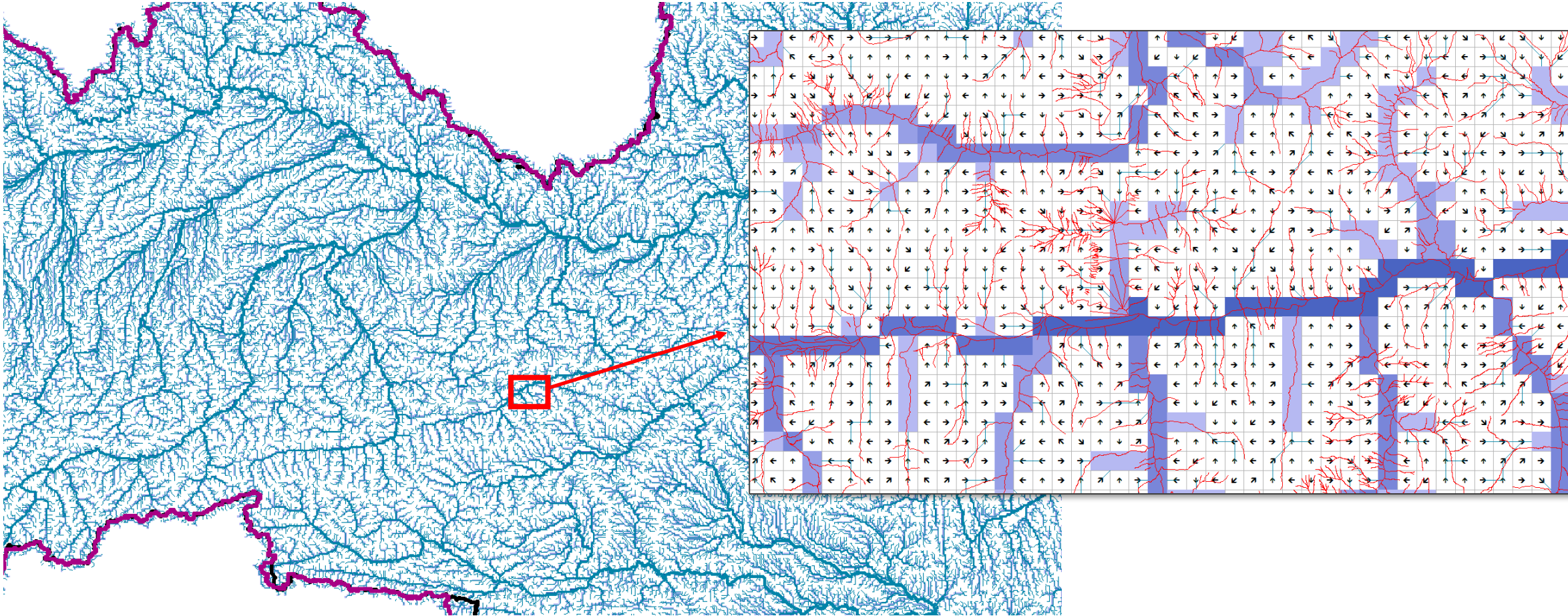
EU_river – shapefile burned into DEM

Manual corrections of the 1 arcmin network



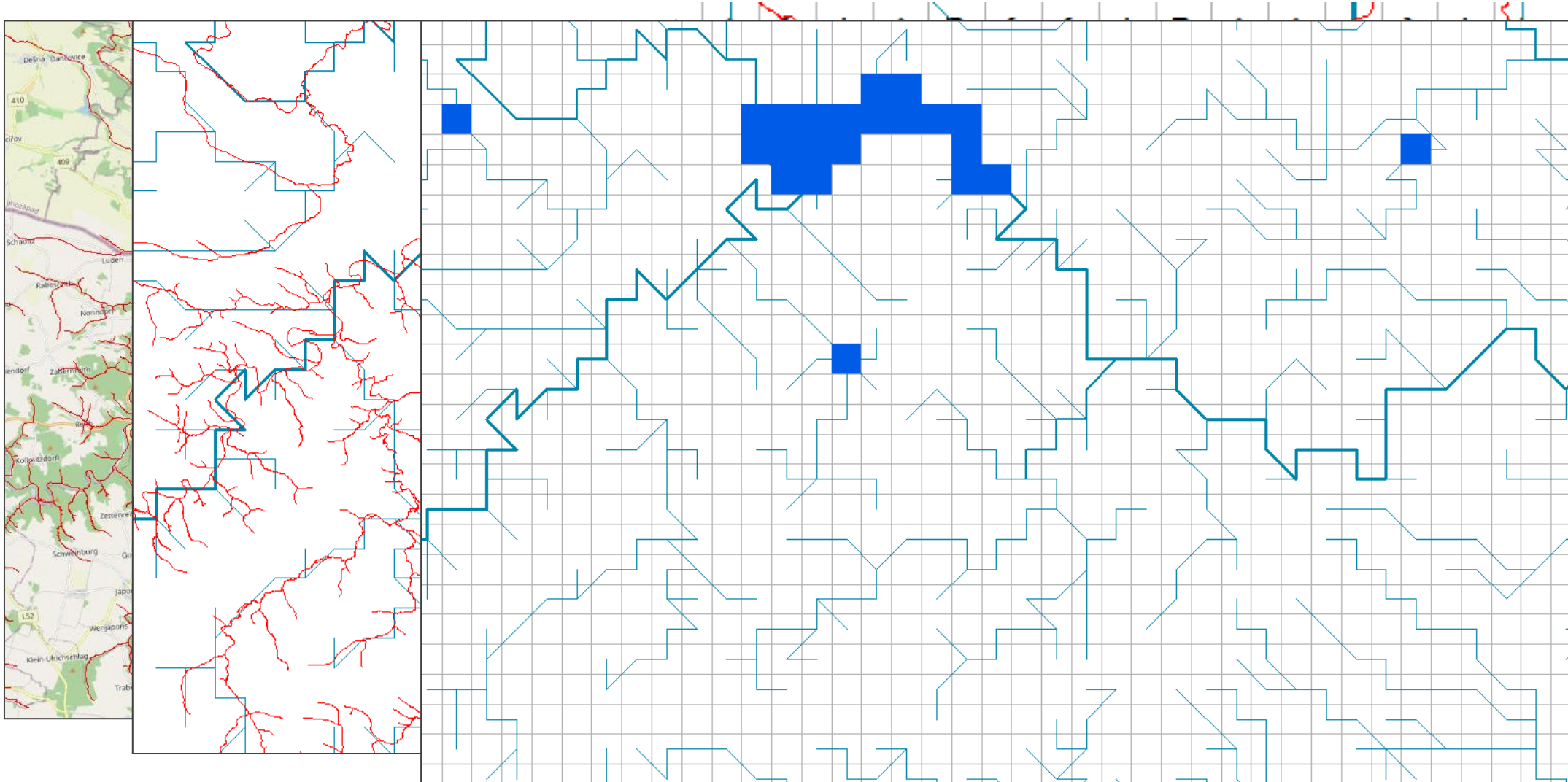
D8 River Network

River network



Using DEM and existing river network maps, but manual correction with country data (e.g. Austria: Gewässernetz)

River network

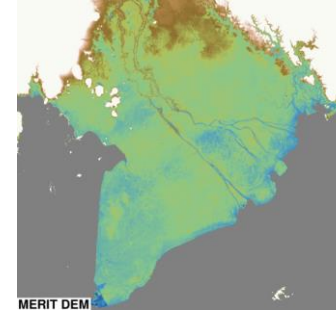


Part of the Thaya – below Vranov dam - Austrian Gewässernetz

River network

Problems:

- D8 Network 1 arcmin vs. real river network
- DEM MERIT and EU-Hydro lead sometimes to wrong Network
- Upscaling from 3 arcsec to 1 armin



https://hydro.iis.u-tokyo.ac.jp/~yamada/MERIT_DEM/



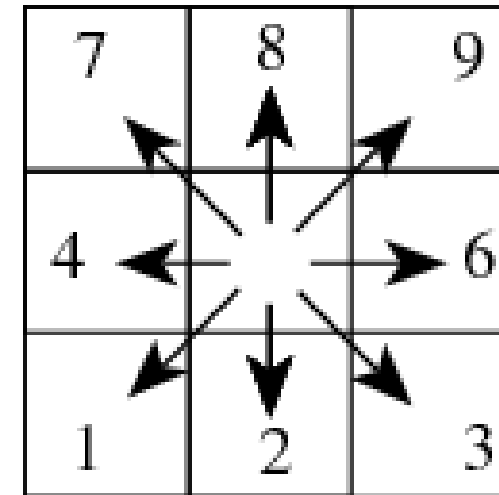
<https://land.copernicus.eu/en/products/eu-hydro>

The need for manual correction of the 1 arcmin network.

- River network as raster map
- Changing single values

ArcGIS Tool:

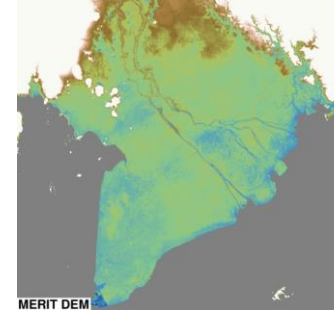
- transforming network raster to point shape
- changing point attribute
- transferring back to rivernetwork, upstream area, ...



River network

Problems:

- D8 Network 1 arcmin vs. real river network
- DEM MERIT and EU-Hydro lead sometimes to wrong Network
- Upscaling from 3 arcsec to 1 armin



https://hydro.iis.u-tokyo.ac.jp/~yamada/MERIT_DEM/



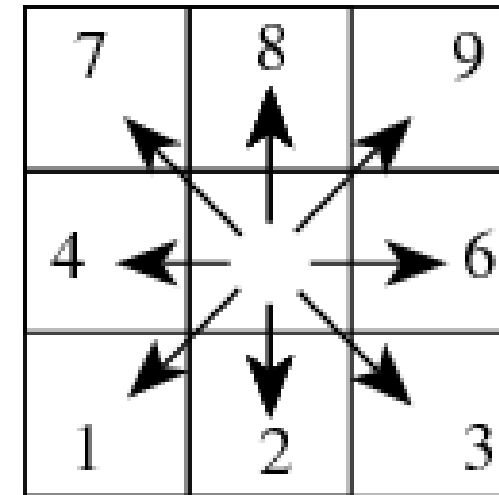
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The need for manual correction of the 1 arcmin network.

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ArcGIS Tool:

- transforming network raster to point shape
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- transferring back to rivernetwork, upstream area, ...



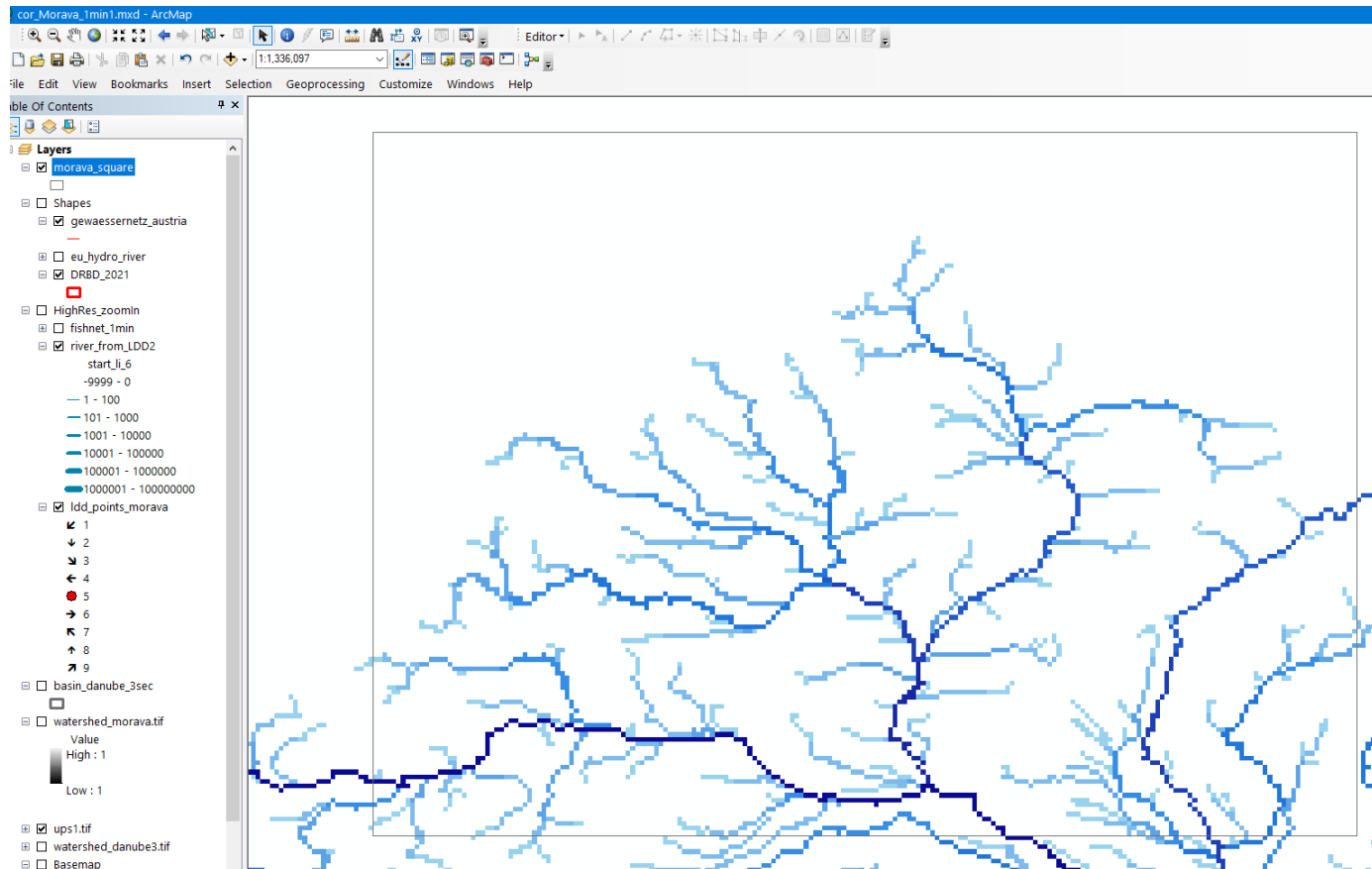
e.g.:

[../Morava/Tool_to_correct_Morava/cor_Morava_1min1.mxd](#)

Tool to correct River network

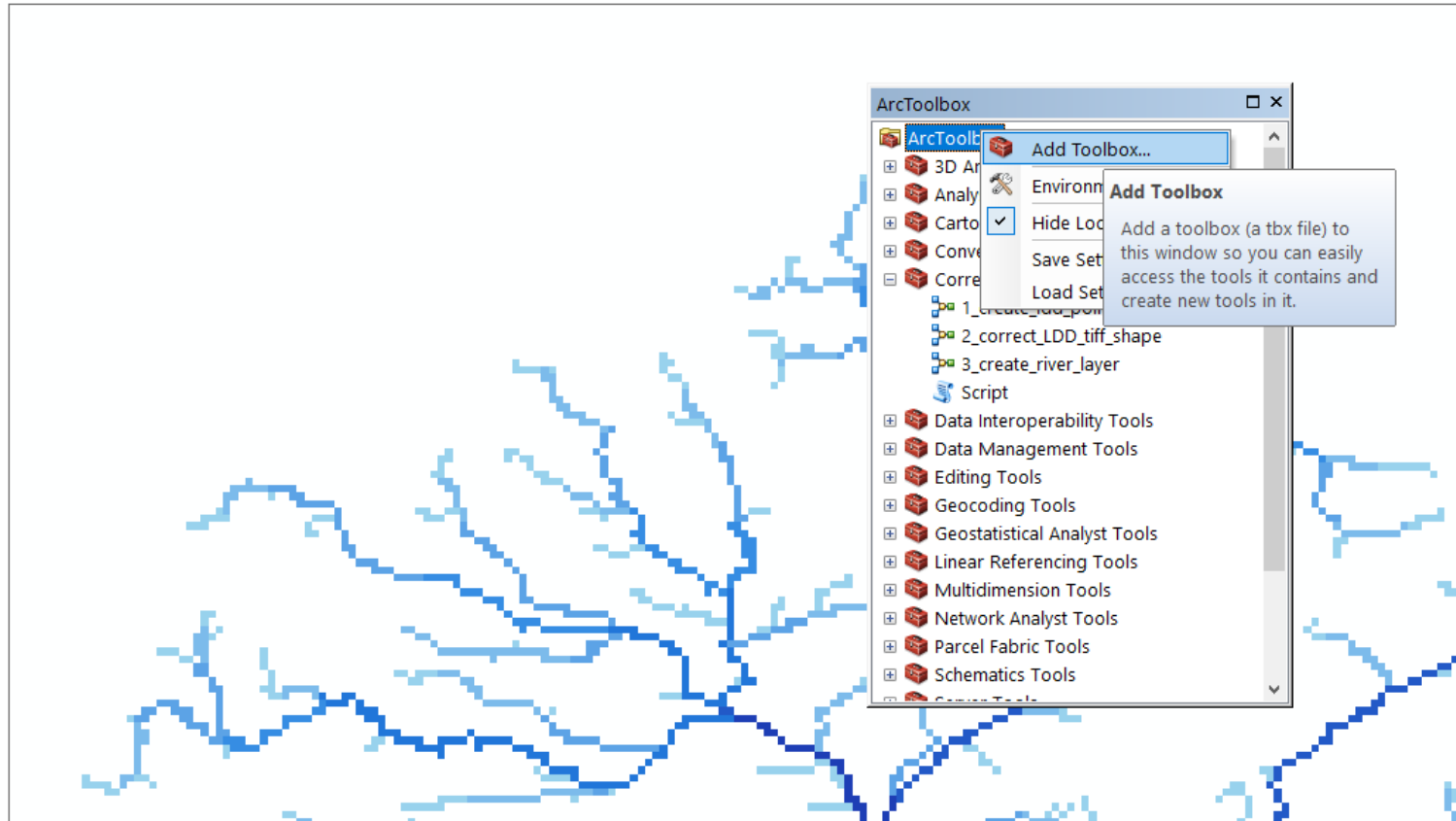
[../Morava/Tool_to_correct_Morava/cor_Morava_1min1.mxd](#)

Documentation: [../Tool_to_correct_Morava/Tool to correct the river network.docx](#)



Tool to correct River network

Step 1: Adding the model builder toolbox Correct_LDD.tbx

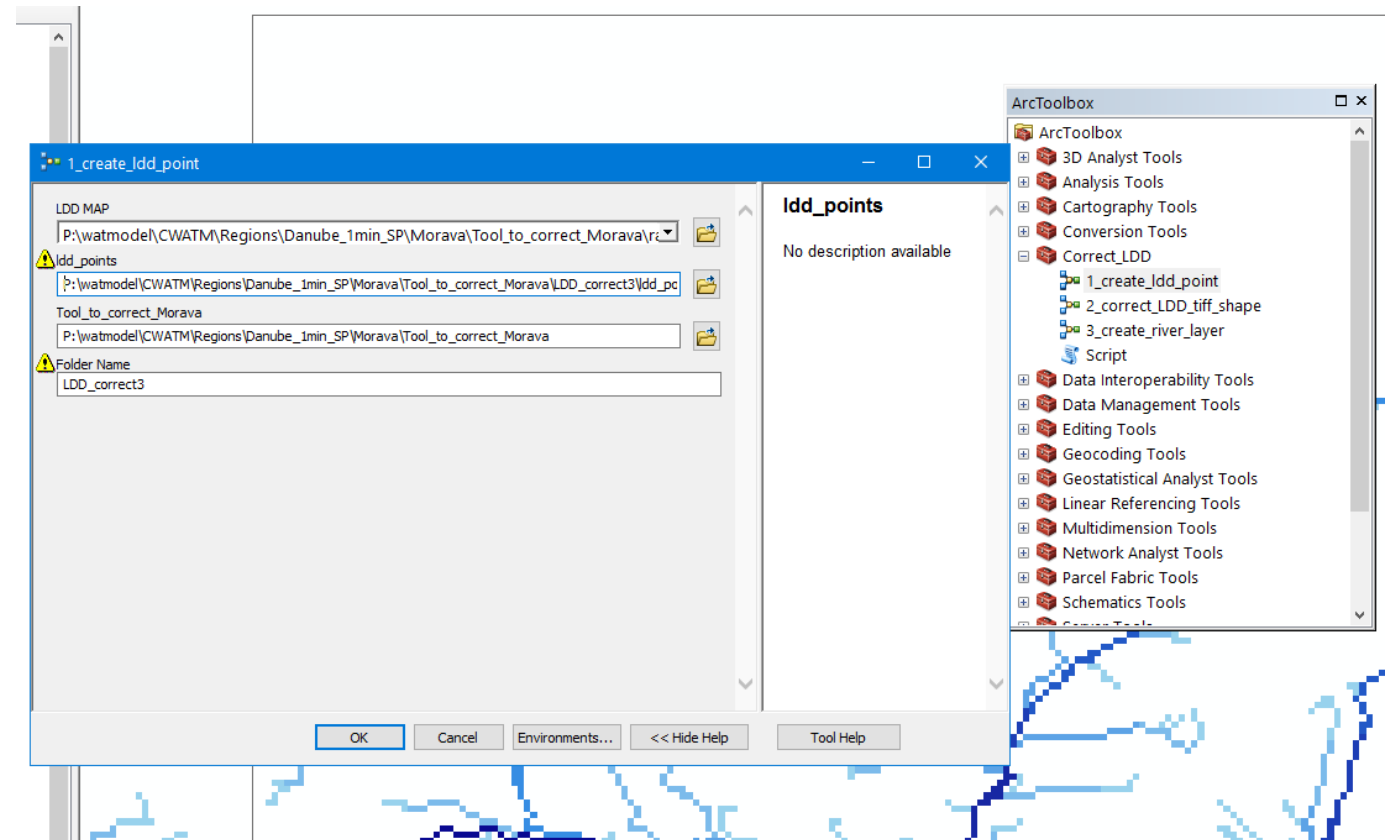


Tool to correct River network

Step 2: Running 1_create_ldd_point

This will use the LDD (river network Raster map pn 1 arcmin) and produce a point shape file

- change location from LDD_correct2 to LDD_correct3
- run the model builder script

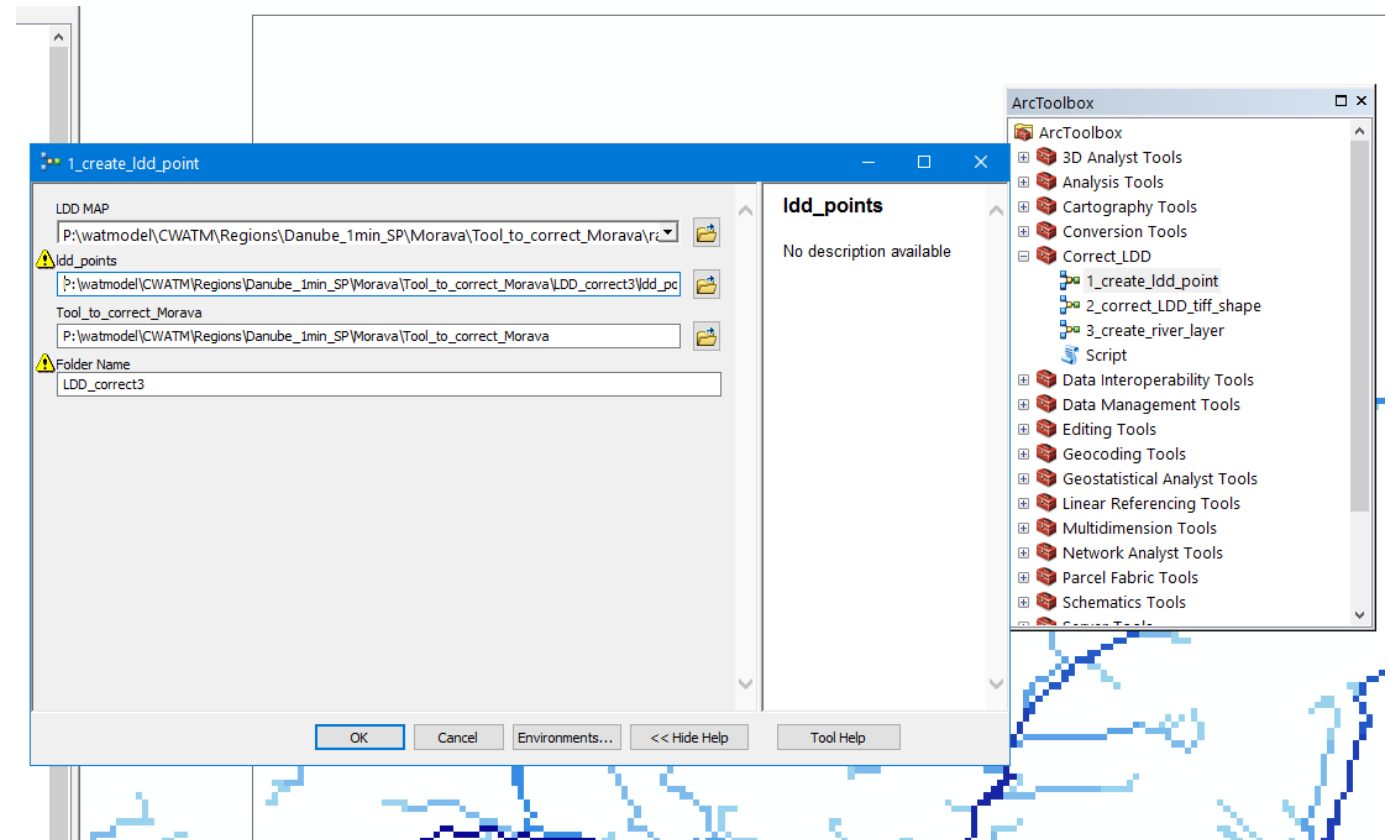


Tool to correct River network

Step 2: Running 1_create_ldd_point

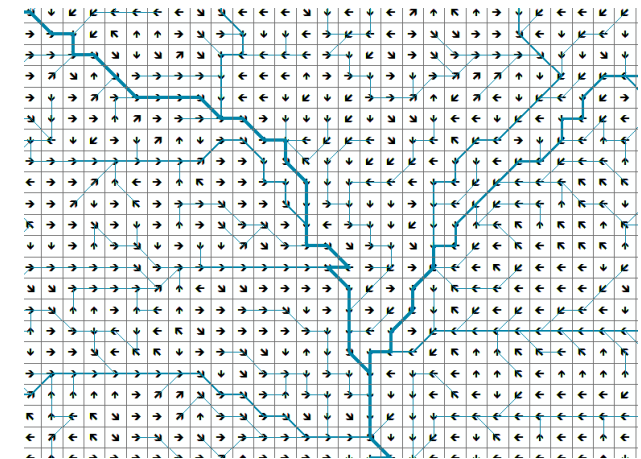
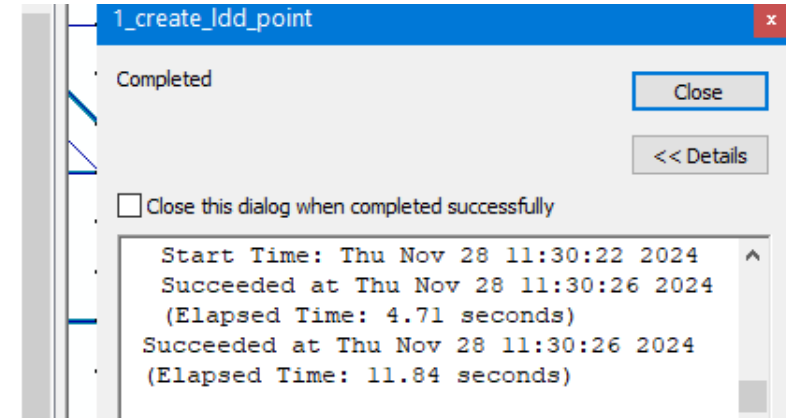
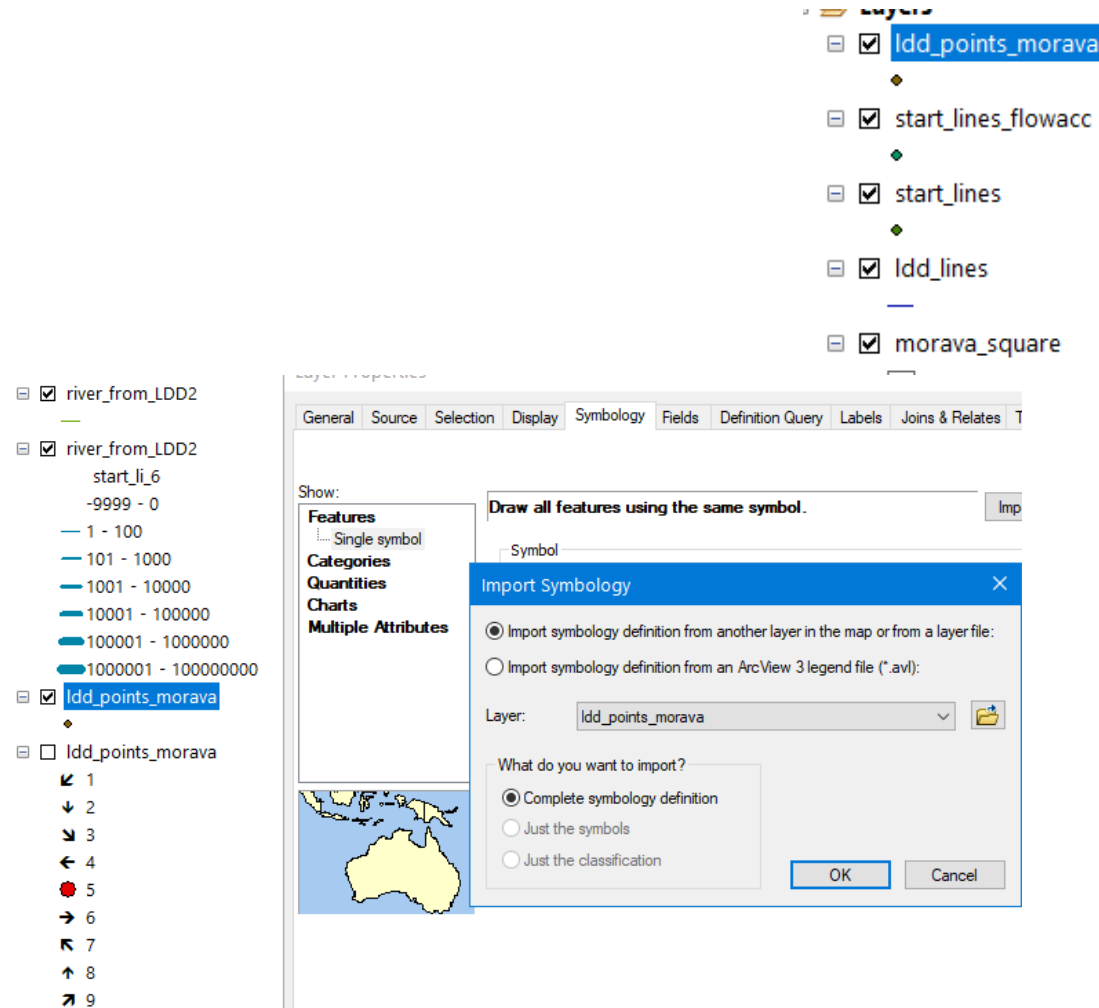
This will use the LDD (river network Raster map pn 1 arcmin) and produce a point shape file

- change location from LDD_correct2 to LDD_correct3
- run the model builder script



Tool to correct River network

Step 3: moving to HighRes, changing symbology, deleting old one



Tool to correct River network

Step 4: Changing point attribute

- make this layer the only selectable
- start editing
- select single point
- change grid code (eg. from 1 (SW) to 2 (South))

The direction is the same as the extended keyboard

7 8 9

4 6

1 2 3

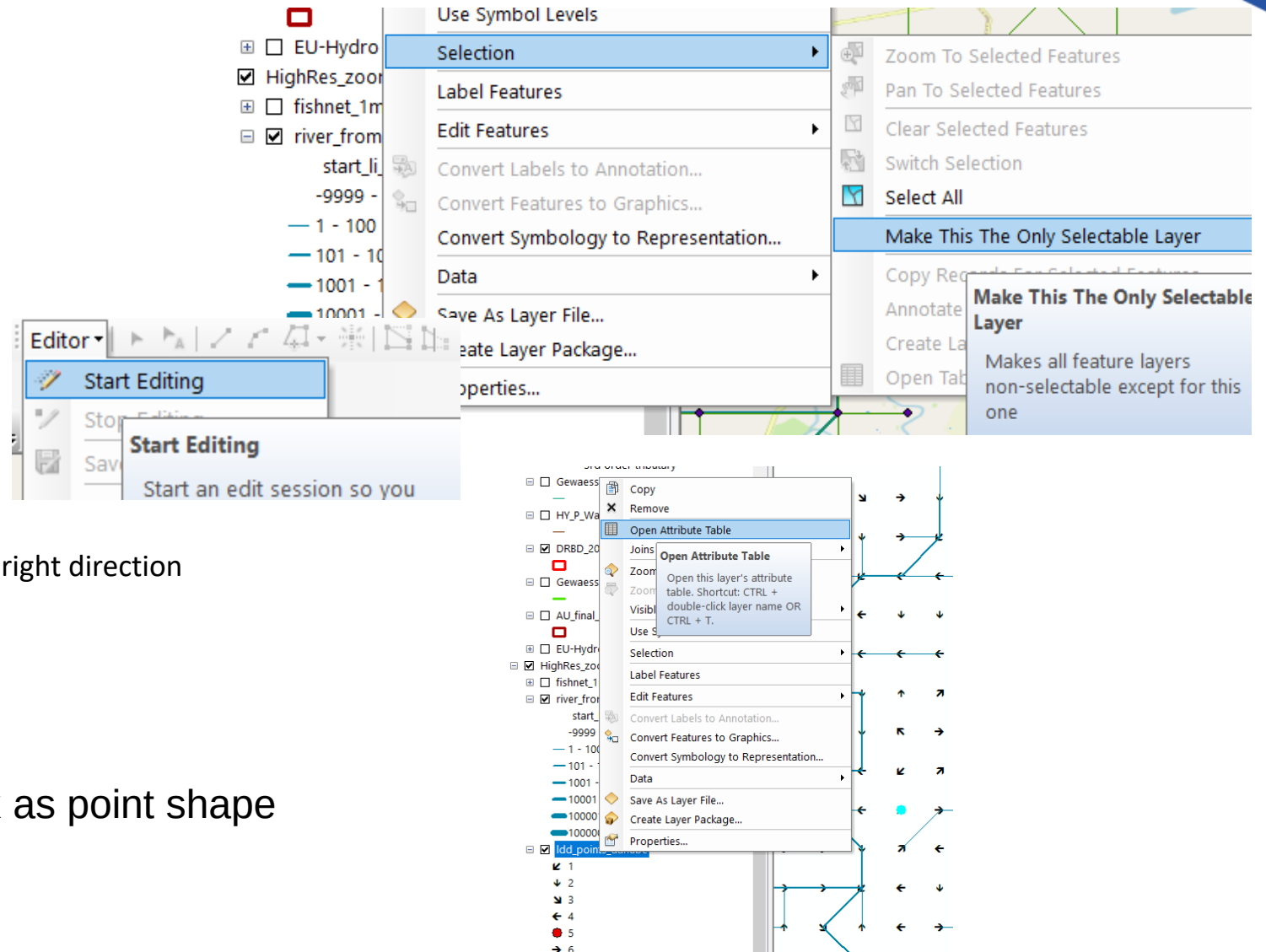
and the 5 is indicating a sink (last point of a basin)

After selecting the next - check if the arrow point in the right direction

- Editor -> save edits

Result:

A better representation of the river network as point shape



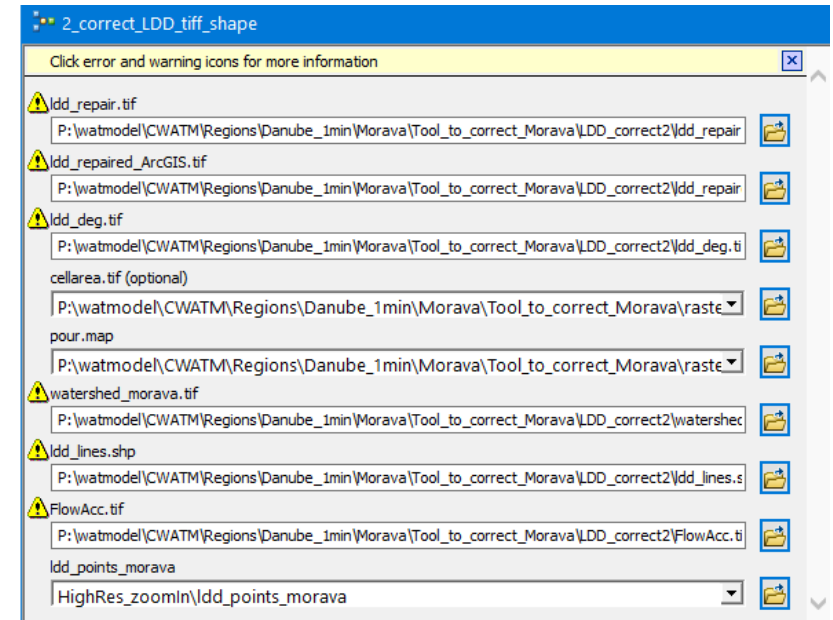
Tool to correct River network

Step 5: Creating a new raster river network (LDD – local drainage direction)

- run 2_correct_LDD_tiff_shape
ArcGis is not overwriting files by default.
If you want to overwrite your files you have to run:
Correct_LDD/Script first
- Change the path in the script

Results:

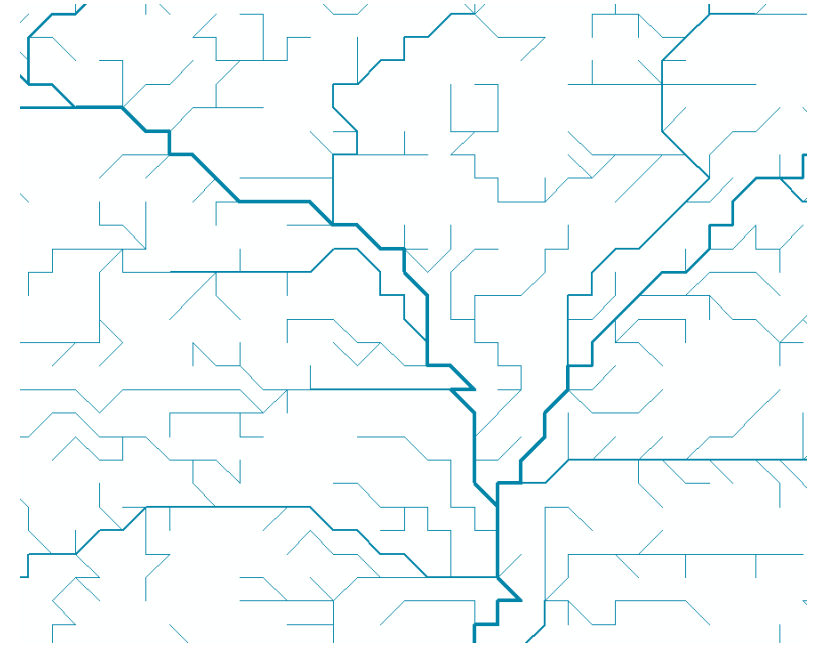
- ldd_repair.tif correct LDD -> used for CWatM
- ldd_repaired_ArcGIS.tif LDD with ArcGis convention of neighbor points
- ldd_deg.tif: LDD with degree as convention neighbor points
e.g. ldd_repair value 7: is pointing to northeast -> ArcGIS: 32 -> degree: 315
- watershed_morava.tif -> watershed of the basin: all points included in the watershed (here you can see, if you excluded some gridcell, by an accidentally circular round)
- ldd_lines.shp : connection lines of grid points
- FlowAcc.tif: upstream area calculated with ArcGis



Tool to correct River network

Step 6: Creating a line shape which shows the direction of the river network

- run 3_create_river_layer
 - change the pathnames
- if you created a new river_from_LDD2
 - move river_from_LDD2 to HighReszoomIn
 - import Symbology

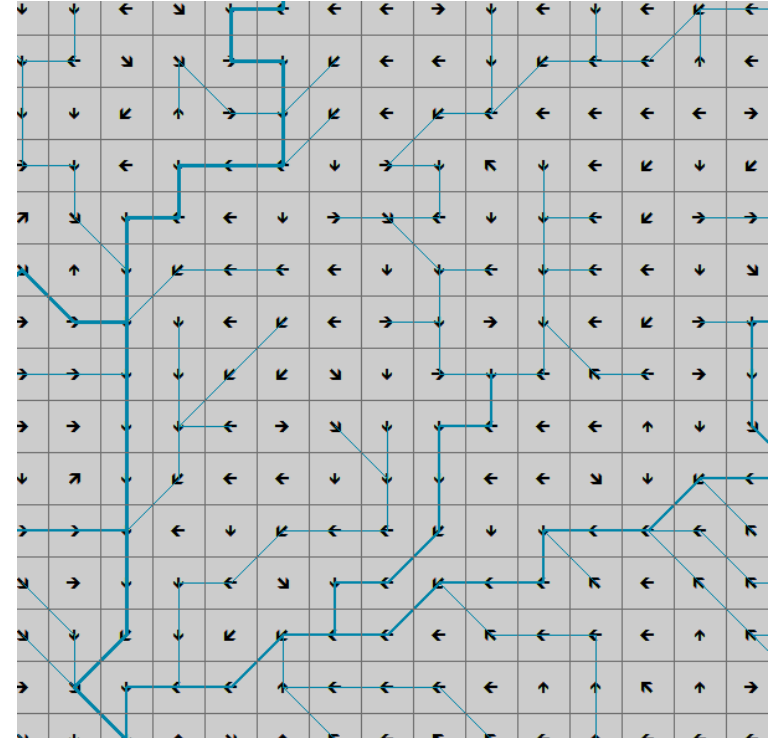
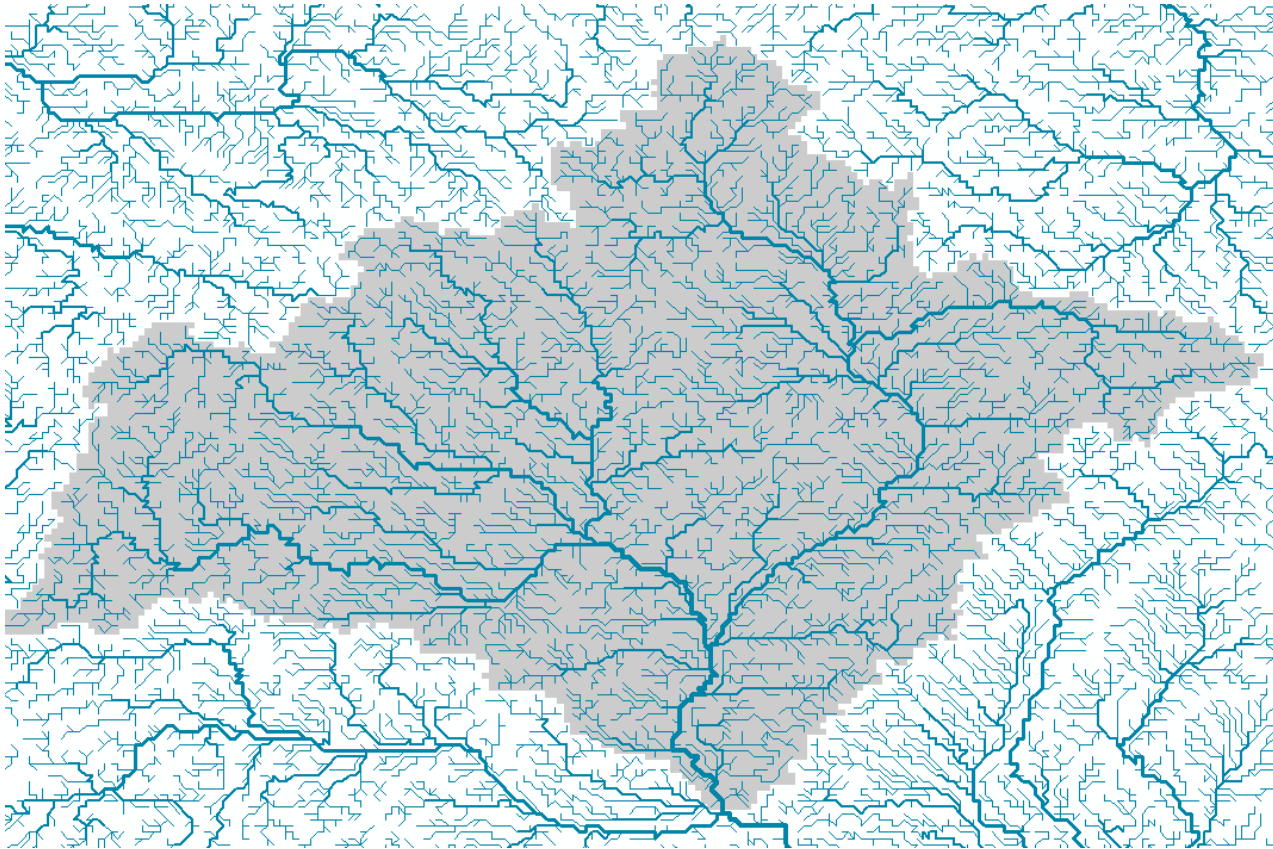


Remarks:

1. try to avoid number <0 and number >9 . This will crash the network.
If you are unsure, that you put a e.g. 23 accidentally in:
sort ldd_points_danube biggest first then lowest first to see if you put in a number >9 or <1
2. try to avoid circular round, e.g. a grid cell is pointing to another grid cell pointing back to the original gridcell (can be bigger loops, too)

Tool to correct River network

Result: River network on 1 arcmin



Tool to correct River network

Next step (next time)

- creating channel maps (chanbw.nc, chanleng.nc, ...) from new ldd_repair.tif
- adjusting lakes and reservoirs with new river network

Please have a look at the documentation:
Tool to correct the river network.docx